
Reproducing and Extending Agent-UniRAG: E5 Reranking, Multi-Hop Decomposition, and Relationship-Aware Cybersecurity Retrieval

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Abstract

Retrieval-augmented generation systems ground a language model’s output in evidence fetched from an external corpus. The quality of that evidence, not the fluency of generation, determines whether a RAG system’s answers can be trusted. Published RAG frameworks are rarely evaluated independently, and their behavior on multi-hop questions and on structured, non-encyclopedic domains such as cybersecurity is not well characterized. This paper reports an independent reproduction and empirical extension of Agent-UniRAG [8], a trainable open-source LLM agent framework for unified single-hop and multi-hop RAG. We keep the original Elasticsearch retriever server and multilingual-E5-large [14] reranker unchanged and evaluate retrieval quality on the full 500-question MuSiQue [12] development set. Dense E5 reranking of the Elasticsearch/BM25 [10] candidate pool reaches Recall@1 of 62.60%, Recall@5 of 82.40%, Recall@10 of 88.20%, and MRR of 71.17%, at 43.29 s per query on a CPU-only workstation. Supplying gold multi-hop reasoning-step decomposition before retrieval and reranking raises these further, to Recall@1 66.80%, Recall@5 90.20%, Recall@10 94.40%, and MRR 76.41%, with 99.60% oracle candidate recall; this decomposition is explicitly not autonomous, since it uses gold reasoning steps rather than a learned planner. A five-question pilot of automatic decomposition with a small local Qwen2.5-0.5B-Instruct [9] model produced malformed, repetitive, and dependency-violating sub-queries, and was judged too unreliable to scale to the full 500 questions; we report this as a negative result rather than omit it. We then extend the UniRAG retrieval interface to a structured cybersecurity corpus built from a pinned MITRE ATT&CK Enterprise STIX 2.1 release (v19.1) [6, 11], indexing 1,898 entity documents and 19,668 explicit relationship documents (21,566 total, 0 indexing errors) after finding that entity-only indexing does not make ATT&CK relationship edges retrievable. On a controlled 10-question pilot of explicit two-hop ATT&CK relationship queries, lexical BM25 retrieval alone already achieves perfect Recall@1 (10/10) on both hops; adding E5 reranking does not improve accuracy and adds roughly 3.2–3.5 s of latency per query. Semantic reranking and multi-hop decomposition therefore behave as task- and corpus-dependent components: valuable for open-domain multi-hop text retrieval, unnecessary overhead for narrow, explicit relationship retrieval over an already-structured knowledge graph. Every number in this paper is drawn directly from a logged experimental run.

1 Introduction

Retrieval-augmented generation has become a standard way to ground a language model’s output in evidence external to its parameters, rather than relying solely on what was learned at pretraining time [5]. Numerous efforts have since explored when to invoke retrieval, how to rerank candidates, and how to decompose complex queries, in pursuit of more reliable knowledge-intensive question answering.

Retrieval quality bounds what such a system can do. No amount of generation-side prompting recovers a candidate set that omits the passage a question actually depends on. Retrieval quality, not generation fluency, is therefore the primary lever for RAG reliability.

Multi-hop questions sharpen this problem. A question such as “which country is the birthplace of the director of the film that won X” cannot be answered from a single passage; each hop’s query depends on an entity resolved by a previous hop, so a single-shot retrieval call against the original question frequently misses intermediate evidence. Two strategies are commonly proposed to address this: *decomposing* the question into an ordered sequence of simpler sub-queries, so that each hop is retrieved with a query that actually mentions the entity it needs [13], and *reranking* a lexical candidate pool with a semantic embedding model, so that paraphrased or non-lexically-overlapping evidence is not lost at the first-stage cutoff [14]. Both strategies add computational cost, and neither is guaranteed to help on every corpus or query type. Whether they are worth that cost is an empirical question, one that depends on corpus structure as much as on method design.

Cybersecurity retrieval is a useful contrast case for this question. Unlike an encyclopedic multi-hop QA corpus, a structured cyber threat intelligence (CTI) knowledge base such as MITRE ATT&CK [11] is organized as an explicit graph of entities (malware, intrusion sets, techniques, mitigations) connected by a small vocabulary of typed relationships (*uses*, *mitigates*). Retrieval over such a corpus can, in principle, be answered by exact or near-exact lexical matching against relationship statements. This raises the question of whether the reranking and decomposition machinery that helps on open-domain multi-hop text is still necessary, or even useful, once the corpus itself already encodes the relevant structure.

In this work, we reproduce Agent-UniRAG [8], a trainable open-source LLM agent framework that unifies single-hop and multi-hop RAG handling in one system, and extend it along three empirical axes built on top of the reproduced retrieval pipeline. This raises seven concrete questions, addressed in turn below: whether the retrieval pipeline reproduces cleanly and can be evaluated independently; whether E5 semantic reranking improves retrieval quality over the Elasticsearch/BM25 candidate pool; how retrieval behaves specifically on hard, four-hop MuSiQue queries; whether decomposition followed by retrieval and reranking improves multi-hop retrieval further; whether the UniRAG retrieval interface can be extended to structured cybersecurity knowledge; how lexical retrieval compares with E5 reranking on structured MITRE ATT&CK relationship retrieval; and whether explicit relationship edges support reproducible multi-hop security retrieval. E5 reranking materially improves retrieval quality on MuSiQue; oracle decomposition improves it further; and, on the cybersecurity side, an explicit relationship-document corpus makes ATT&CK graph edges retrievable at all, though lexical matching alone turns out to be sufficient once that corpus exists. Throughout, we distinguish between original Agent-UniRAG functionality we reused unmodified, our own reproduction and extensions, oracle/controlled experiments that use privileged information not available to an autonomous system, and negative results, which we report with the same rigor as positive ones. In the sections that follow, we describe the reproduced pipeline and each extension in turn, then report results and discuss what they imply about when reranking and decomposition are worth their cost.

2 Background and Related Work

Retrieval-augmented generation combines a parametric generator with a non-parametric retrieval index, originally formulated by Lewis et al. [5] for knowledge-intensive NLP tasks. Subsequent work has explored when and how to invoke retrieval, how to rerank candidates, and how to decompose complex queries; this paper is concerned specifically with the retrieval stage rather than end-to-end generation quality.

Agent-UniRAG [8] proposes a trainable LLM-agent controller with Planning, Tool-Use, Working-Memory, and Reflector modules that decides, step by step, whether a query needs single-hop or

multi-hop handling, and fine-tunes an open-source backbone on a synthetic SynAgent-RAG dataset to imitate this behavior end-to-end. Its official implementation reuses an Elasticsearch retriever server and a multilingual-E5-large dense reranker, and evaluates on MuSiQue, HotpotQA, and 2WikiMultihopQA alongside single-hop benchmarks, following corpus preparation from IRCot [13]. This project reuses that retriever server and reranker architecture without modification to the original agent-controller logic; our contribution is an independent empirical evaluation of the retrieval stage in isolation, plus new extensions, not a reproduction of the trained agent-controller itself.

MuSiQue [12] constructs genuinely multi-hop 2–4-hop questions bottom-up from composed single-hop questions specifically to resist single-hop shortcut solutions. IRCot [13] interleaves chain-of-thought generation with retrieval so that each reasoning step’s retrieval query is conditioned on what has already been derived. A broader family of decomposition approaches similarly breaks a complex query into an ordered sequence of sub-queries that are resolved sequentially. A recent study [2] pairs LLM-driven question decomposition with an off-the-shelf cross-encoder reranker and reports retrieval and answer-accuracy gains on multi-hop benchmarks, consistent with the pattern we observe under an oracle decomposition setting in Section 5, though our decomposition source differs and our numbers are not directly comparable.

Multilingual E5 [14] is a family of text embedding models trained with weakly-supervised contrastive pre-training followed by supervised fine-tuning, initialized from XLM-RoBERTa-large for the large variant used here. As a reranker it is applied downstream of a first-stage lexical retriever (BM25 [10] via Elasticsearch), rather than as a stand-alone dense retriever, which is the configuration used by Agent-UniRAG and preserved in this work.

MITRE ATT&CK [11] is a knowledge base of adversary tactics, techniques, and procedures derived from real-world observations, distributed as machine-readable STIX 2.1 [4] bundles via the official `attack-stix-data` repository [6]. Prior work has explored RAG over MITRE ATT&CK for cyber threat intelligence: RAGIntel [1] builds a hybrid BM25/dense retrieval system with reranking and compression over an HTML-derived ATT&CK knowledge base for attack-investigation question answering, evaluated on 339 queries across three benchmarks. Our cybersecurity extension differs in scope: rather than optimizing end-to-end attack-investigation QA, we focus narrowly on whether a UniRAG-style retrieval interface can retrieve explicit graph relationship edges from a STIX-derived corpus, and we compare lexical and dense-reranked retrieval on that narrow task.

3 Method

3.1 Retrieval Infrastructure Reproduction

We reproduced the retrieval infrastructure described in the official Agent-UniRAG repository [7] rather than replacing it: an Elasticsearch 7.10.2 index [3] serving as the first-stage lexical retriever, exposed through a FastAPI-based retriever server (`/retrieve/` endpoint) on port 8000, with paragraph-level MuSiQue passages as the indexed unit and multilingual-E5-large as an optional downstream dense reranker. This is original Agent-UniRAG functionality reused as-is; our contribution at this stage is standing the infrastructure up independently and validating it end-to-end. The reproduction was carried out in a local Windows/Conda environment; a recurring Windows SSL certificate-store issue affected `aiohttp/elasticsearch-client` TLS imports during retriever-server startup, which we worked around locally. We note this only as an environment caveat, not a research contribution.

3.2 E5 Reranking

For each query–candidate pair returned by Elasticsearch, the reranker (i) encodes the query and candidate with multilingual-E5-large, (ii) L2-normalizes both embeddings, (iii) scores the pair by cosine similarity, and (iv) sorts candidates in descending score order. Because both embeddings are normalized, cosine similarity reduces to a dot product:

$$\text{score}(q, d) = \frac{E_q \cdot E_d}{\|E_q\| \|E_d\|} = E_q \cdot E_d, \quad \|E_q\| = \|E_d\| = 1. \quad (1)$$

Figure 1 (left) illustrates this pipeline. The reranking logic is our own implementation built around the pre-trained, unmodified E5 weights.

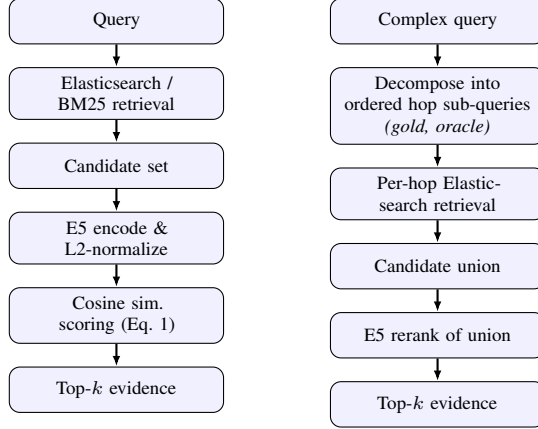


Figure 1: (left) E5 reranking pipeline. (right) Multi-hop decomposition pipeline, using gold hop sub-queries in the controlled oracle experiment of Section 5.2.

3.3 Multi-Hop Decomposition

To evaluate whether decomposition helps beyond reranking alone, we built a decomposition-then-retrieve-then-rerank pipeline (Figure 1, right). For each question, the pipeline uses an ordered sequence of hop sub-queries to issue one retrieval call per hop, unions the resulting candidates across hops, and reranks the union with E5. In the controlled experiment reported in Section 5, the hop sub-queries are the question’s *gold* MuSiQue reasoning-step decomposition — privileged information not available to an autonomous system at inference time. Section 5 also reports what happened when we replaced this oracle with a small local automatic planner.

3.4 Cybersecurity Corpus Construction

We selected MITRE ATT&CK Enterprise as a structured cybersecurity corpus and downloaded a pinned STIX 2.1 snapshot, `enterprise-attack-19.1.json`, from the official `attack-stix-data` repository. The raw dataset contained 25,843 STIX objects; after filtering deprecated and revoked objects and selecting the relevant entity classes, 1,898 entity documents remained (Table 6). We initially indexed only these entity documents. Testing revealed a structural limitation: entity documents alone do not make ATT&CK relationship edges (e.g. “which technique does this malware use”) directly retrievable, since no single entity document expresses a relationship. We therefore added 18,220 *uses* and 1,448 *mitigates* relationship documents (19,668 total), each stating an explicit relationship such as “APT38 uses Scheduled Task” or “Network Intrusion Prevention mitigates Symmetric Cryptography.” This relationship-document design change is the central methodological contribution of the cybersecurity extension. All 21,566 documents were indexed into Elasticsearch with 0 errors, and direct retrieval confirmed relationship statements are searchable, e.g. “3PARA RAT uses Symmetric Cryptography” was returned correctly for a query about that malware’s techniques. The UniRAG `/retrieve/` API was connected to this corpus using the same interface validated in Section 3.1; we do not claim the resulting system is an autonomous cybersecurity reasoning agent, only a retrieval-oriented extension with relationship-aware documents.

4 Experimental Setup

Environment. Windows, Conda environment `agent-unirag`, transformers 4.46.3, torch 2.4.1+cpu, Elasticsearch 7.10.2 on port 9200, UniRAG retriever API on port 8000. All latency numbers reported below are specific to this CPU-only, single-machine setup and are not claimed to generalize to GPU deployments or other hardware.

Datasets and benchmarks. MuSiQue evaluation used the full 500-question set from the reproduced pipeline. A 19-case four-hop diagnostic subset was evaluated separately, at two different points in the project under different pipeline configurations, to isolate whether hard four-hop failures come from candidate generation, multi-hop structure, or reranking. The cybersecurity benchmark went

Table 1: MuSiQue E5 reranking results.

Evaluated	Recall@1	Recall@5	Recall@10	MRR	Latency
100 (checkpoint)	0.5800	0.8100	0.8700	0.6792	54.0501 s
500 (final)	0.6260	0.8240	0.8820	0.7117	43.2863 s

Table 2: Four-hop diagnostic evaluation (19 cases each; separate runs, not a paired comparison).

Run	Gold cand. rec.	Recall@1	Recall@5	Recall@10	MRR	Latency
Early E5 reranking	not measured	0.3158	0.4211	0.4211	0.3684	36.1980 s
Decomposition-oriented	1.0000	0.4211	0.5789	0.6316	0.4971	27.7315 s

through one corrective iteration: an initial 10-question benchmark generated directly from ATT&CK relationships had repeated source→technique pairs, repeated near-identical questions, and technique names leaked directly into the question text; a corrected generator enforced unique pairs, unique question text, 3-hop chains, no name leakage, and deterministic, structurally validated selection, producing the final 10-question benchmark used for all reported cybersecurity results. We describe this benchmark throughout as a controlled pilot, not a statistically powered benchmark.

Automatic planner pilot. We tested automatic question decomposition with a locally cached Qwen2.5-0.5B-Instruct model on 5 questions, to assess whether the oracle-decomposition gains could be realized without gold supervision.

Metrics. For a ranked list returned for query q , Recall@ k is the fraction of queries for which the gold evidence appears in the top k results:

$$\text{Recall@}k = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}[\text{gold}_q \in \text{Top-}k(q)], \quad (2)$$

and mean reciprocal rank is

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q}, \quad (3)$$

with $1/\text{rank}_q = 0$ if the gold evidence is absent from the ranked list. *Candidate recall* (or *gold candidate recall*) is the fraction of queries for which the gold evidence appears anywhere in the pre-rerank candidate pool, used in Section 5 to separate candidate-generation failures from reranking failures. *Latency* is average wall-clock time per query in the environment above.

5 Results

5.1 MuSiQue Retrieval

Table 1 reports E5 reranking results on MuSiQue. The 500-question row is final; the 100-question row is an earlier checkpoint and is not used as a final metric.

On the full 500-question set, dense reranking reaches Recall@1 of 62.60%, Recall@5 of 82.40%, Recall@10 of 88.20%, and MRR of 71.17%. A lexical-only (BM25, no rerank) baseline on this same 500-question set was not separately measured in this experiment and is not reported as a number here.

Table 2 reports the 19-case four-hop diagnostic, run twice under different configurations (rows are separate runs, not a before/after pair). The decomposition-oriented run’s perfect gold candidate recall (1.0000) against a much lower Recall@1 (0.4211) indicates the gold passage was consistently present in the candidate pool but not always ranked first — on this small subset, late-stage ranking, not early-stage candidate generation, is the binding constraint.

Table 3 profiles the CPU-only reranker. Increasing batch size from 8 to 32 increased latency (42.456 s → 57.439 s) rather than reducing it — batching did not produce the speedup typically expected on GPU hardware, on this particular setup. Latency increased substantially with sequence length up to 384 tokens (31.559 s → 41.455 s) and then plateaued (384 → 512: 41.455 s → 41.659 s), suggesting most candidates were already shorter than 384 tokens. A separate real-query profiling run showed per-query latency varying substantially, with some queries exceeding 100 s.

Table 3: E5 CPU reranker profiling (multilingual-e5-large, CPU-only).

Batch size (100 cand.)	Latency	Seq. length	Latency
8	42.456 s	128	31.559 s
16	47.949 s	256	37.583 s
32	57.439 s	384	41.455 s
		512	41.659 s

Table 4: Oracle decomposition + E5 reranking, all 500 MuSiQue questions, vs. plain E5 reranking on the same questions.

Metric	E5 only	Decomp. + E5	Δ
Recall@1	0.6260	0.6680	+0.0420
Recall@5	0.8240	0.9020	+0.0780
Recall@10	0.8820	0.9440	+0.0620
MRR	0.7117	0.7641	+0.0524
Gold candidate recall	—	0.9960	—

5.2 Multi-Hop Decomposition

Table 4 reports the controlled oracle-decomposition experiment on the same 500 MuSiQue questions. Every metric improves over plain E5 reranking (Table 1, 500-question row): Recall@1 by 4.2 points, Recall@5 by 7.8 points, Recall@10 by 6.2 points, and MRR by 5.2 points, at essentially unchanged latency (42.32 s vs. 43.29 s). Gold candidate recall of 99.60% indicates that once per-hop retrieval is guided by gold decomposition, the correct evidence is almost always present in the unioned pool; the remaining gap to perfect Recall@10 is attributable to reranking, mirroring the four-hop diagnostic pattern above. We emphasize again that this experiment uses gold reasoning-step decomposition and upper-bounds, rather than measures, autonomous performance.

5.3 Automatic Decomposition Pilot

The Qwen2.5-0.5B-Instruct planner’s outputs were inconsistent across the five-question pilot: it sometimes generated sensible ordered sub-queries, but it also produced malformed or free-form output instead of the requested structured format, repeated the same query across hops, jumped directly to an attempted final answer instead of decomposing, used unresolved placeholders for entities not yet known, and in some cases generated a later-hop query that referenced information not yet resolved. In one representative case, the planner generated a query about the final-answer entity while the intermediate entities needed to reach it were still unknown. Because the pilot is small (N=5) and failures were logged qualitatively, we report observed failure categories rather than fabricated per-category counts (Table 5).

This is a documented negative result: the small local planner was not judged reliable enough to serve as the decomposition source for an autonomous 500-question benchmark, and we did not scale it up. It qualifies Section 5.2 directly — the oracle result demonstrates the *potential* of decomposition, while this pilot shows that realizing that potential with a 0.5B-parameter local model, under the prompting scheme used here, is not yet reliable.

5.4 Cybersecurity Relationship Retrieval

Table 6 summarizes the final corpus. A dedicated evaluator, `evaluate_cyber_hops.py`, tests two explicit relationship hops directly: source entity $\rightarrow$ attack technique (e.g. “3PARA RAT $\rightarrow$ Symmetric Cryptography”) and mitigation $\rightarrow$ attack technique (e.g. “Network Intrusion Prevention $\rightarrow$ Symmetric Cryptography”). This evaluator replaced an initial end-to-end benchmark evaluation that had produced misleading near-zero scores due to a string-matching bug between entity names, document titles, and API metadata; those discarded numbers are not used anywhere as a performance claim and are restated only in the Appendix for transparency.

On this controlled benchmark, BM25 alone achieves the correct relationship document at rank 1 in all 10/10 cases on both hops, with sub-50-millisecond latency. Adding E5 reranking does not change

Table 5: Automatic Qwen2.5-0.5B-Instruct decomposition pilot: observed failure categories (N=5, qualitative).

Observed failure category
Malformed / free-form output instead of structured sub-queries
Repeated the same sub-query across hops
Jumped directly to a final-answer attempt instead of decomposing
Used unresolved placeholders for entities not yet known
Generated a later-hop query violating the sequential dependency structure

Table 6: Cybersecurity corpus composition (MITRE ATT&CK Enterprise v19.1, filtered).

Entity class	Count
attack-pattern	697
malware	726
intrusion-set	174
x-mitre-data-component	106
tool	95
campaign	56
course-of-action	44
Entity documents	1,898
uses relationships	18,220
mitigates relationships	1,448
Relationship documents	19,668
Total indexed	21,566
Indexing errors	0

accuracy (still 10/10 at rank 1) but increases latency by roughly two orders of magnitude. Structured ATT&CK relationship retrieval was highly effective with lexical retrieval alone; dense E5 reranking provided no measured accuracy gain on this 10-question pilot and incurred substantial additional latency. We do not claim this generalizes beyond the scope of this controlled pilot, nor that E5 is universally worse than BM25.

6 Discussion

Table 8 consolidates the primary findings. E5 reranking improves retrieval quality on MuSiQue across every metric, and oracle decomposition improves it further, with the largest gain in Recall@5. Candidate recall under oracle decomposition is extremely high (99.60%), so the residual gap to perfect Recall@10 is a reranking problem, not a candidate-generation problem — a pattern that also appears in the smaller four-hop diagnostic. Automatic decomposition with a small local model is not yet reliable, based on a five-question pilot; this is a genuine negative result rather than an implementation bug, since the model ran correctly but produced structurally invalid or dependency-violating output. CPU-only E5 reranking is expensive in this environment, and batching does not help. Structured ATT&CK relationship retrieval is highly effective with BM25 alone, and E5 reranking does not improve, and substantially slows, that narrow task.

Taken together, these results draw a line between two retrieval regimes. For generic, semantic open-domain text retrieval, where the same fact can be expressed with substantial lexical variation across paragraphs, as in MuSiQue, dense reranking measurably helps, and multi-hop decomposition helps further when decomposition is reliable. For explicit structured relationship retrieval over a corpus that already encodes graph edges as short, low-lexical-variation statements, as in the MITRE ATT&CK relationship-document corpus, lexical matching can already be sufficient, and the added computational cost of dense reranking is not justified by the accuracy evidence gathered here. Dense embeddings appear to earn their cost primarily by bridging lexical gaps between query and evidence phrasing; when a corpus is engineered so that relationship statements already lexically overlap with natural relationship queries, as our “X uses Y” / “X mitigates Y” documents do by construction, that gap is largely closed before reranking is invoked.

Table 7: Cybersecurity relationship-hop evaluation, 10 questions.

Hop	Retriever	Recall@1	Recall@5	Recall@10	MRR	Latency
1: source→technique	BM25	1.0000	1.0000	1.0000	1.0000	0.0406 s
	BM25+E5	1.0000	1.0000	1.0000	1.0000	3.2122 s
2: mitigation→technique	BM25	1.0000	1.0000	1.0000	1.0000	0.0390 s
	BM25+E5	1.0000	1.0000	1.0000	1.0000	3.4638 s

Table 8: Summary of primary results.

Finding	Result
E5 reranking, 500-Q MuSiQue	R@1 62.60%, R@10 88.20%, MRR 71.17%
Oracle decomposition + E5, 500-Q MuSiQue	R@1 66.80%, R@10 94.40%, MRR 76.41%, cand. recall 99.60%
Automatic 0.5B planner pilot	Unreliable on 5-Q pilot; not scaled
Cyber corpus	21,566 docs (1,898 entity + 19,668 relationship), 0 index errors
Cyber relationship retrieval	BM25 100% R@1 (10/10, both hops); BM25+E5 same accuracy, $\sim 80\times$ latency

Several scope limitations bound these conclusions. The cybersecurity relationship-hop benchmark and the automatic-planner pilot are both small (10 and 5 questions respectively) and should be read as controlled pilots, sensitive to the specific entities and questions selected, not statistically powered studies. The oracle decomposition result is an upper bound on autonomous performance, not a measurement of it. All CPU timings are specific to the Windows/Conda, CPU-only environment used throughout and are not claimed to generalize to GPU hardware. The cybersecurity corpus reflects MITRE ATT&CK’s Enterprise domain as of the pinned v19.1 snapshot, not the full space of cybersecurity knowledge, and no user study or real security-operations-center deployment was performed.

7 Conclusion

In this work, we presented an independent reproduction of Agent-UniRAG’s retrieval infrastructure, evaluated it at scale on MuSiQue, and extended it with a controlled multi-hop decomposition experiment, a documented negative-result pilot for automatic decomposition, and a relationship-aware MITRE ATT&CK cybersecurity retrieval corpus.

On the full 500-question MuSiQue set, E5 reranking improved retrieval quality over the underlying candidate pool, reaching Recall@1 of 62.60% and MRR of 71.17%. Oracle multi-hop decomposition improved these further, to Recall@1 66.80% and MRR 76.41%, with 99.60% gold candidate recall. A small local automatic decomposition planner was not reliable enough to realize these gains without gold supervision, based on a five-question pilot.

A relationship-aware MITRE ATT&CK corpus of 21,566 documents made explicit cybersecurity graph-edge retrieval possible in the first place. On a controlled 10-question relationship benchmark, BM25 alone already retrieved the correct edge at rank 1 in every case; E5 reranking added roughly two orders of magnitude of latency without improving accuracy.

Taken together, these results argue for treating semantic reranking and multi-hop decomposition as task- and corpus-dependent components, applied where lexical overlap between query and evidence is weak, rather than as universally beneficial modules. We plan to evaluate a larger, better-constrained automatic decomposition planner against the oracle upper bound established here, and to expand the cybersecurity relationship benchmark beyond its current 10-question pilot scope. The retriever server, evaluation scripts, and MITRE ATT&CK corpus-construction pipeline used in this work are documented in full in the accompanying repository.

References

- [1] Abeer Alhuzali. LLM-powered threat intelligence: A retrieval-augmented generation approach for cyber attack investigation. *PeerJ Computer Science*, 11:e3371, 2025.
- [2] Paul J. L. Ammann and Jonas Golde. Question decomposition for retrieval-augmented generation, 2025.
- [3] Elastic NV. Elasticsearch 7.10.2. <https://www.elastic.co/downloads/past-releases/elasticsearch-7-10-2>. Search engine used by the original Agent-UniRAG retriever server.
- [4] Bret Jordan, Rich Piazza, and Trey Darley. STIX version 2.1. <https://docs.oasis-open.org/cti/stix/v2.1/stix-v2.1.html>, 2021.
- [5] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.
- [6] MITRE ATT&CK. attack-stix-data: STIX data representing MITRE ATT&CK. <https://github.com/mitre-attack/attack-stix-data>. Enterprise ATT&CK release v19.1, accessed August 2026.
- [7] Hoang Pham. Agent-unirag official code repository. <https://github.com/pvhoang14/Agent-UniRAG>, 2025. Accessed August 2026.
- [8] Hoang Pham, Thuy-Duong Nguyen, and Khac-Hoai Nam Bui. Agent-unirag: A trainable open-source llm agent framework for unified retrieval-augmented generation systems, 2025.
- [9] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- [10] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009.
- [11] Blake E. Strom, Andy Applebaum, Doug P. Miller, Kathryn C. Nickels, Adam G. Pennington, and Cody B. Thomas. MITRE ATT&CK: Design and philosophy. Technical report, The MITRE Corporation, 2020. Originally published July 2018, revised March 2020.
- [12] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. MuSiQue: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022.
- [13] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10014–10037, Toronto, Canada, 2023. Association for Computational Linguistics.
- [14] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. Multilingual E5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*, 2024.

Appendix: Discarded Initial Cybersecurity Evaluation

Table 9 restates, for transparency, the initial end-to-end cybersecurity evaluation numbers discussed in Section 5.4. These predate the dedicated relationship-hop evaluator and were affected by an evaluator string-matching bug; they are superseded entirely by Table 7 and should not be used as a performance claim.

Table 9: Discarded initial end-to-end cybersecurity evaluation (evaluator string-matching bug; not a final result).

Configuration	Recall@1	Recall@5	Recall@10	MRR
BM25	0	0	0	0
BM25+E5 (corrected run)	not measured	not measured	≈ 0.10	not measured
Oracle-MultiHop+E5 (corrected run)	not measured	not measured	≈ 0.10	not measured